① **Support of fuzzy set:-** The support of fuzzy set $\tilde{A}$ on X is a classical set which is defined by

$$\text{Support}(\tilde{A}) = \{ x \in X : \mu_A(x) > 0 \}$$

eg:- Let $X = \{1, 2, 3\}$

$$\tilde{A} = \{(1, \cdot 2), (2, \cdot 3), (3, 0)\}$$

$$\text{Support}(\tilde{A}) = \{1, 2\}$$

② **α-cut set or cut worthy set:-**

The α-cut set of fuzzy set $\tilde{A}$ of the set X is a crisp set which is defined

$$^{\alpha}A = \{ x \in X : \mu_{\tilde{A}}(x) \geq \alpha \}$$ (which is crisp set)

~~eg:~~ where $\alpha \in [0, 1]$

eg:- Let $X = \{1, 2, 3, 4\}$,

$$\tilde{A} = \{(1, \cdot 2), (2, \cdot 4), (3, \cdot 6), (4, \cdot 9)\}$$

Find α-cut set for $\alpha = \cdot 3$

Solution:-

$$^{\cdot 3}\tilde{A} = \{2, 3, 4\}, \text{ where } \alpha = \cdot 3$$

Since,

$$\mu_{\tilde{A}}(1) = \cdot 2 < 3$$

$$\mu_{\tilde{A}}(2) = \cdot 4 > 3$$

$$\mu_{\tilde{A}}(3) = \cdot 6 > 3$$

$$\mu_{\tilde{A}}(4) = \cdot 9 > 3$$

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③ Strong α-cut set :-

The strong α-cut set of fuzzy set $\tilde{A}$ of the set X is a crisp set which is defined by

$$\alpha^{+}A = \{x \in X : \mu_A(x) > \alpha\}, \text{ where } \alpha \in [0,1]$$

EX. Let $X = \{1, 2, 3\}$,

$$\tilde{A} = \{(1, .2), (2, .4), (3, .6), (4, .9)\}$$

~~find~~ $.3^{+}\tilde{A} = \{2, 3, 4\}$

$.2^{+}\tilde{A} = \{2, 3, 4\}$

$.4^{+}\tilde{A} = \{3, 4\}$

④ Convex fuzzy set :- A fuzzy set $\tilde{A}$ is said to be convex iff $\mu_{\tilde{A}}(\lambda x_1 + (1-\lambda)x_2)$

$$\geq \min\{\mu_{\tilde{A}}(x_1), \mu_{\tilde{A}}(x_2)\}$$

where $\lambda \in [0,1]$

⑤ Decomposition of fuzzy set

$$_{\alpha}A = \alpha^{\alpha}A$$

⑥ Cardinality of fuzzy set :-

Cardinality of fuzzy set is defined as

$$|\tilde{A}| = \sum \mu_{\tilde{A}}(x)$$

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⑥ Relative cardinality :-

Relative cardinality is defined as $Re(\tilde{A}) = \frac{|\tilde{A}|}{10}$

Ex. Let $X = \{1, 2, 3, 4\}$

$\tilde{A} = \{(1, .2), (2, .4), (3, .6), (4, .9)\}$

find $|\tilde{A}|$ and $Re|\tilde{A}|$.

Solution:- we know,

$$|\tilde{A}| = \sum \mu_{\tilde{A}}(x) = .2 + .4 + .6 + .9$$
$$= 2.1$$

and $Re|\tilde{A}| = \frac{|\tilde{A}|}{10} = \frac{2.1}{10} = 0.21$

Note:- ① $0 < Re|\tilde{A}| < 1$

Height of a Fuzzy set :-

The height $h(\tilde{A})$ of a fuzzy set $\tilde{A}$ is the largest membership grade obtained by any element in that set i.e.

$$h(\tilde{A}) = \sup_{x \in X} \tilde{A}(x)$$

eg:- $\tilde{A} = \{(1, 0.1), (2, 0.2), (3, 0.9)\}$

$\therefore h(\tilde{A}) = (3, 0.9)$

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Ex. If X is a non empty set such that $X = \{1, 2, 3, 4\}$ and $\bar{A} = \{ \frac{1}{.2}, \frac{2}{.4}, \frac{3}{.6}, \frac{4}{.9} \}$ and $\bar{B} = \{ \frac{1}{.3}, \frac{2}{.5}, \frac{3}{.7}, \frac{4}{1} \}$ are two fuzzy sets on X then find the following

(a) $\bar{A} \cup \bar{B}$ (b) $\bar{A} \cap \bar{B}$ (c) height of $\bar{A}$ and $\bar{B}$

(d) $\alpha_{\bar{A}}$ when $\alpha = .4$ (e) Support of $\bar{A}$ and $\bar{B}$

Solution:

$$\cancel{\bar{A} \cup \bar{B} = \{ \frac{1}{.2}, \frac{2}{.4}, \frac{3}{.6}, \frac{4}{.9} \}}$$

$$\cancel{\bar{A} \cap \bar{B} = \{ \frac{1}{.3}, \frac{2}{.5}, \frac{3}{.7}, \frac{4}{1} \}}$$

$$\bar{A} \cup \bar{B} = \{ \frac{1}{.3}, \frac{2}{.5}, \frac{3}{.7}, \frac{4}{1} \}$$

$$\bar{A} \cap \bar{B} = \{ \frac{1}{.2}, \frac{1}{.4}, \frac{3}{.6}, \frac{4}{.9} \}$$

height of $\bar{A} = \frac{4}{.9}$

height of $\bar{B} = \frac{4}{1}$

Support of $\bar{A} = \{1, 2, 3, 4\}$

Support of $\bar{B} = \{1, 2, 3, 4\}$

$$\alpha_{\bar{A}} = {}^{.4}\bar{A} = \{2, 3, 4\}$$

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